

Problem Statement: Optimizing RAG Document Retrieval for Agronomic Advice

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Smallholder farmers and agricultural extension workers in West Africa often struggle to find timely, trustworthy and locally relevant guidance on crops, pests, diseases, soils, fertilisers and climate adaptation. Farmers may describe symptoms in local, informal or conversational terms, while extension documents use technical language. Consequently, existing retrieval tools can rank keyword-similar but irrelevant documents above information addressing the farmer's actual crop and condition, leading to misleading advice, wasted resources and possible harm. Our project will improve how relevant extension documents are identified and ranked while being guided by fairness, safety, privacy, transparency, accountability and cultural inclusion. We will compare performance with TF-IDF using nDCG@K, particularly nDCG@5, and involve agricultural experts in judging relevance. Success means more dependable access to appropriate evidence, fewer misleading recommendations, protected farm data and better-informed agricultural decisions without replacing qualified extension workers.